

# Shikhar Rai

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 Patna, India

## SUMMARY

Electronics and Communication Engineering student at BIT Mesra with a strong interest in backend development and scalable systems. I have hands-on experience building full-stack applications using **Java and Spring Boot**, and I am currently exploring **Industrial IoT protocols** to improve wireless communication efficiency. I enjoy solving practical problems and actively contribute to both technical projects and team activities.

## EDUCATION

|             |                                                                                                                   |                                   |
|-------------|-------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| 2023 – 2027 | <b>B.Tech in Electronics and Communication Engineering</b><br>Birla Institute of Technology, Mesra (Patna Campus) | <b>CGPA: 7.54</b><br>Patna, India |
| 2022        | <b>Class 12th (ISC Board)</b> — City Montessori School (CMS)                                                      | <b>89%</b>                        |
| 2020        | <b>Class 10th (CBSE Board)</b> — RLB Memorial School                                                              | <b>94%</b>                        |

## TECHNICAL EXPERTISE

|                             |                                                                                                                 |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>Software Development</b> | Experience in building backend logic using <b>Java (OOP)</b> and developing REST APIs with <b>Spring Boot</b> . |
| <b>Database Management</b>  | Skilled in designing relational databases using <b>MySQL</b> and writing efficient, optimized SQL queries.      |
| <b>Tools &amp; Workflow</b> | Familiar with <b>Git, Maven</b> , IntelliJ IDEA, and VS Code for development and version control.               |
| <b>Core Knowledge</b>       | Understanding of <b>Wireless Sensor Networks (WSN)</b> , VLSI Design, and Industrial Electronics.               |

## PROJECTS

### Enhanced MAC Protocol for IIoT — Minor Research Project Feb 2026 – Apr 2026

- Designed an optimized **MAC protocol** to improve data reliability and energy efficiency in Industrial IoT environments.
- Analyzed **IEEE research papers** to connect theoretical WSN concepts with real-world industrial challenges.
- Proposed a framework for simulation and future implementation of the protocol.

### V Shobha's Boutique E-Commerce Platform — Java, Spring Boot, MySQL Jan 2026 – Mar 2026

- Developed a **full-stack e-commerce platform** to help an offline boutique establish an online presence.
- Built backend features for **secure image uploads** and product validation.
- Implemented RESTful APIs and search functionality for efficient product retrieval.

### Expense Tracker — Java, JDBC, MySQL Sept 2025 – Nov 2025

- Developed a **personal expense tracking application** to manage and categorize daily expenses.
- Improved database query performance, achieving approximately **30% faster data processing**.

## ACHIEVEMENTS & LEADERSHIP

- Selected as a **Smart India Hackathon Finalist** for building impactful technology solutions.
- Served as **Captain of the ECE Basketball Team** for two consecutive years, leading team coordination and performance.
- Completed **NPTEL certifications** in Industrial Electronics and Economics.
- Contributed as a **Technika Coordinator**, managing event coordination and sponsorship outreach.